

A: Datasheet

Algorithm: qnap_005

Developer: Qnap Security

Submission Date: 2023_11_16

Template size: 2048 bytes

Template time (2.5 percentile): 959 msec

Template time (median): 972 msec

Template time (97.5 percentile): 985 msec

Investigation:

Frontal mugshot ranking 211 (out of 457) -- FNIR(1600000, 0, 1) = 0.0032 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 266 (out of 419) -- FNIR(1600000, 0, 1) = 0.0255 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 98 (out of 388) -- FNIR(1600000, 0, 1) = 0.1011 vs. lowest 0.0517 from sensetime_009

Immigration visa-border ranking 203 (out of 346) -- FNIR(1600000, 0, 1) = 0.0073 vs. lowest 0.0006 from cloudwalk_mt_002

Immigration visa-kiosk ranking 124 (out of 290) -- FNIR(1600000, 0, 1) = 0.0783 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 124 (out of 457) -- FNIR(1600000, T, L+1) = 0.0102, FPIR=0.001000 vs. lowest 0.0011 from idemia_010

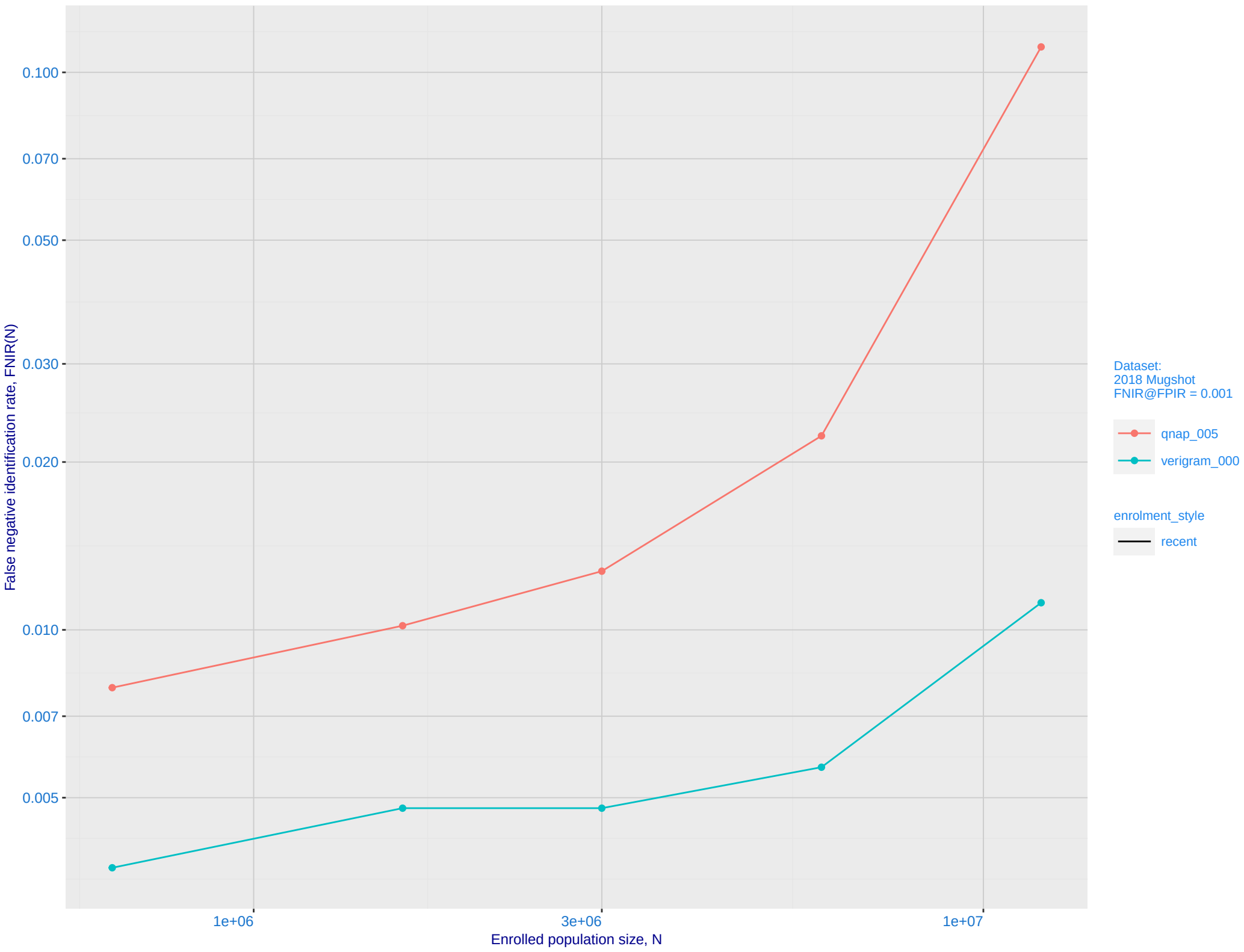
Mugshot webcam ranking 156 (out of 417) -- FNIR(1600000, T, L+1) = 0.0656, FPIR=0.001000 vs. lowest 0.0072 from sensetime_009

Mugshot profile ranking 49 (out of 387) -- FNIR(1600000, T, L+1) = 0.3678, FPIR=0.001000 vs. lowest 0.0634 from cloudwalk_mt_002

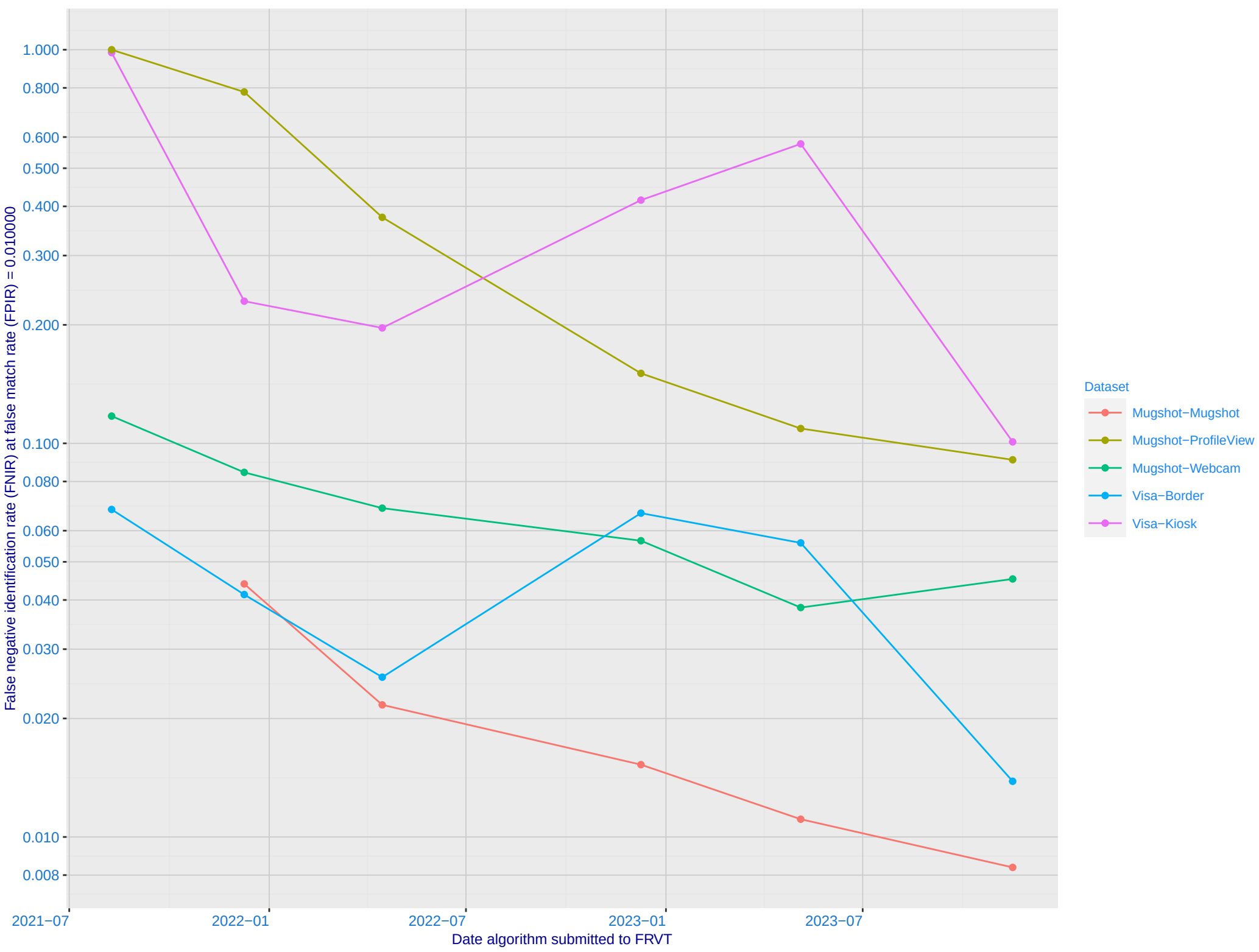
Immigration visa-border ranking 132 (out of 343) -- FNIR(1600000, T, L+1) = 0.0238, FPIR=0.001000 vs. lowest 0.0010 from cloudwalk_mt_002

Immigration visa-kiosk ranking 73 (out of 288) -- FNIR(1600000, T, L+1) = 0.1198, FPIR=0.001000 vs. lowest 0.0517 from cloudwalk_mt_002

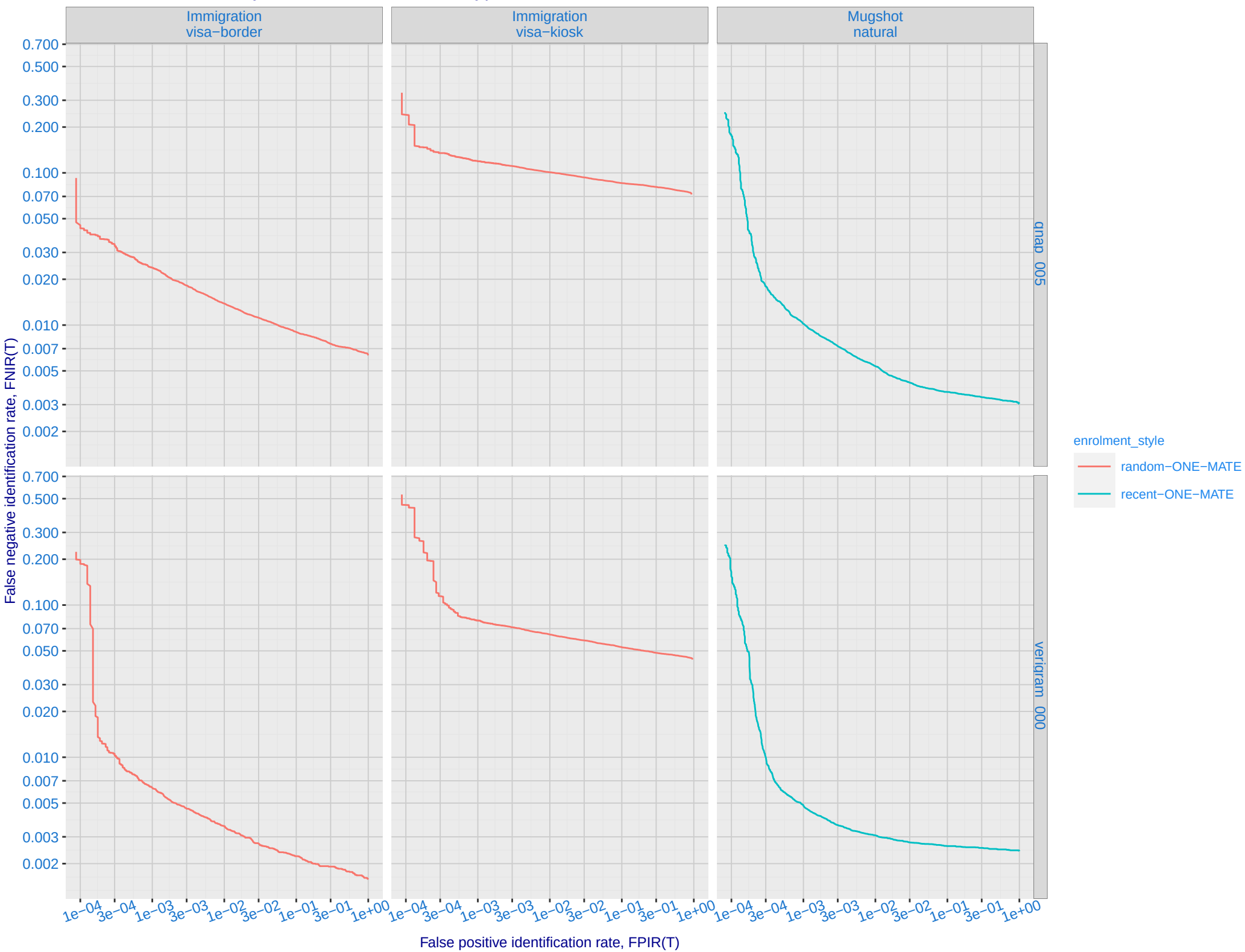
B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (verigram_000)



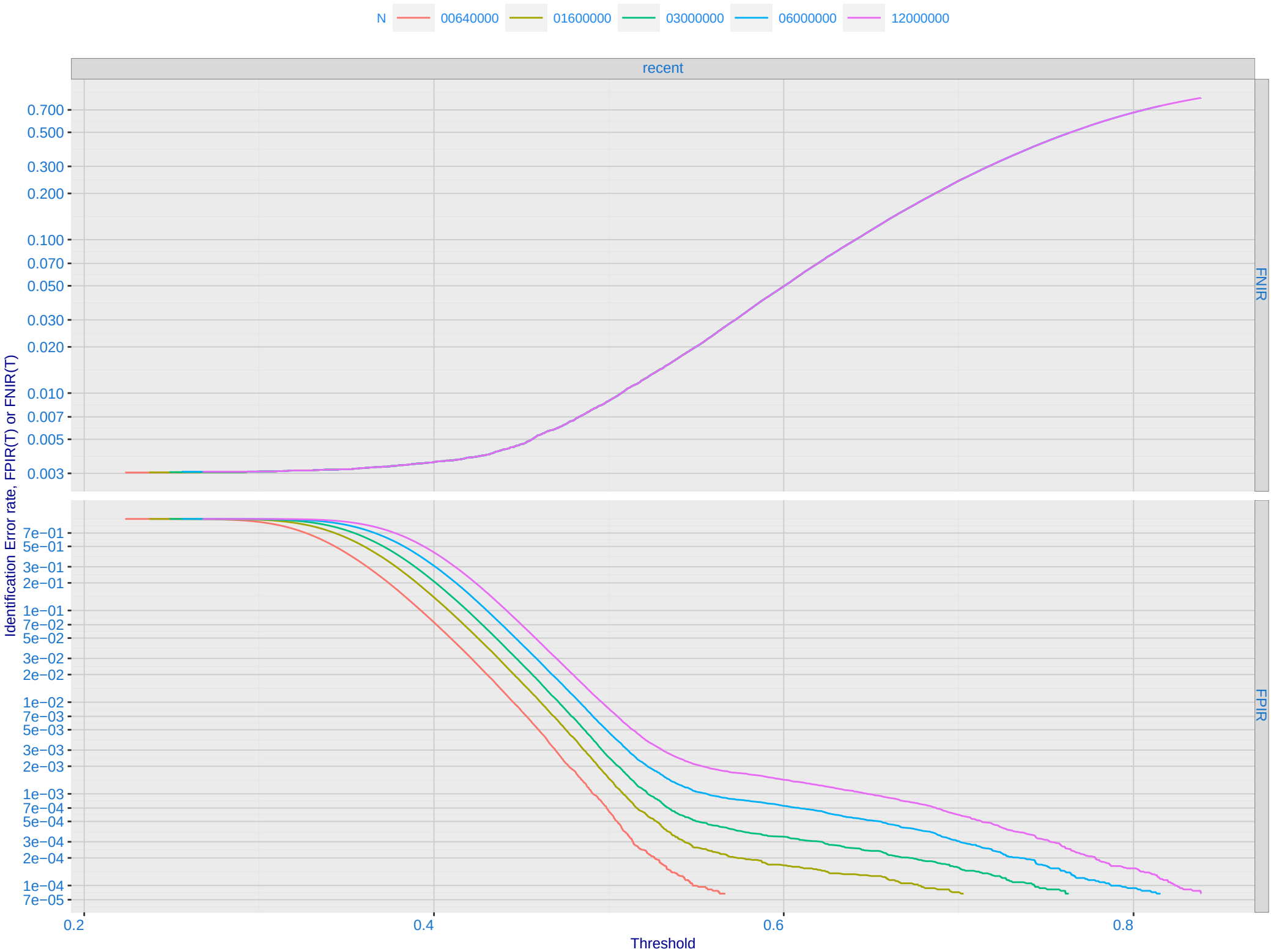
C: Evolution of accuracy for QNAP algorithms on three datasets 2018 – present



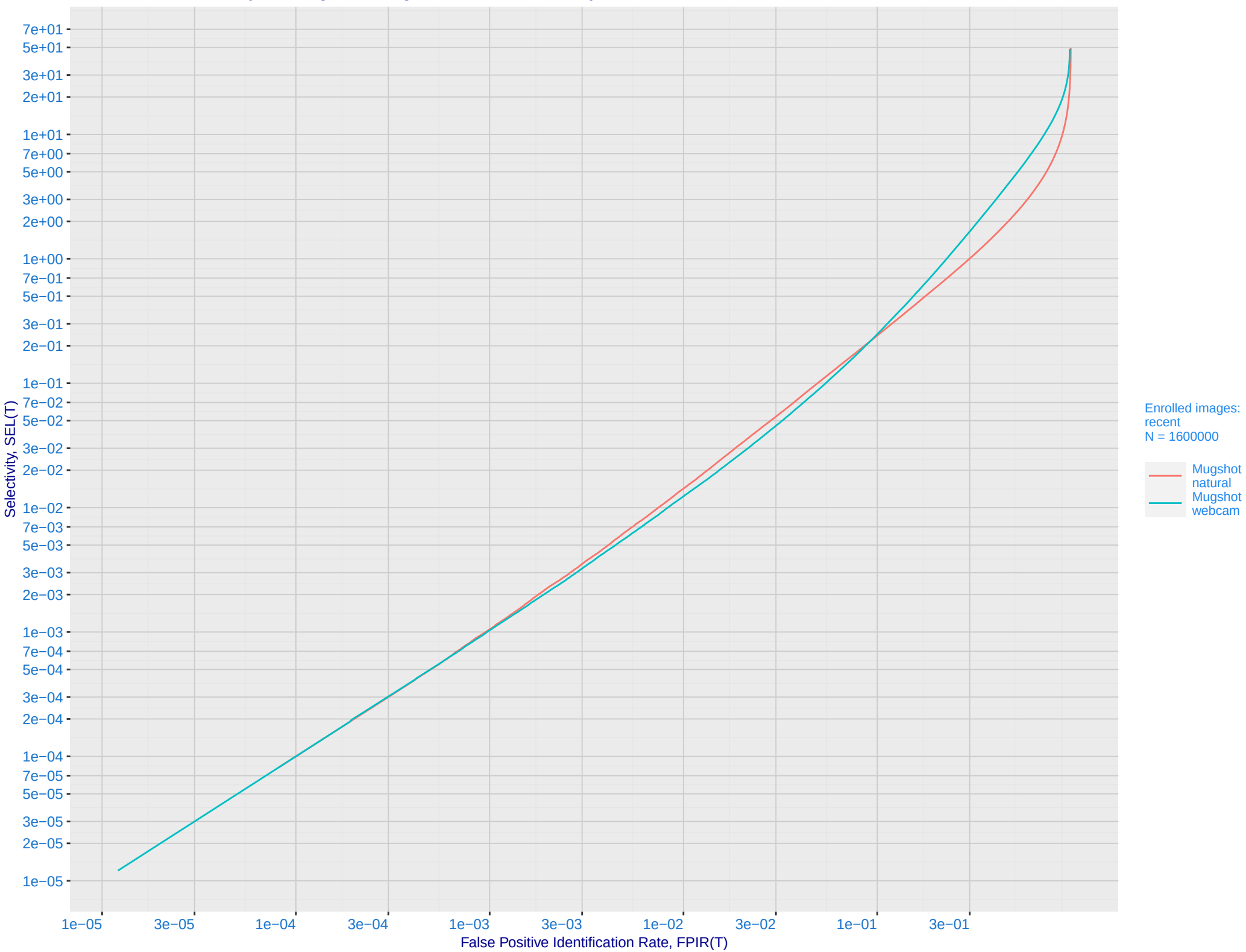
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals



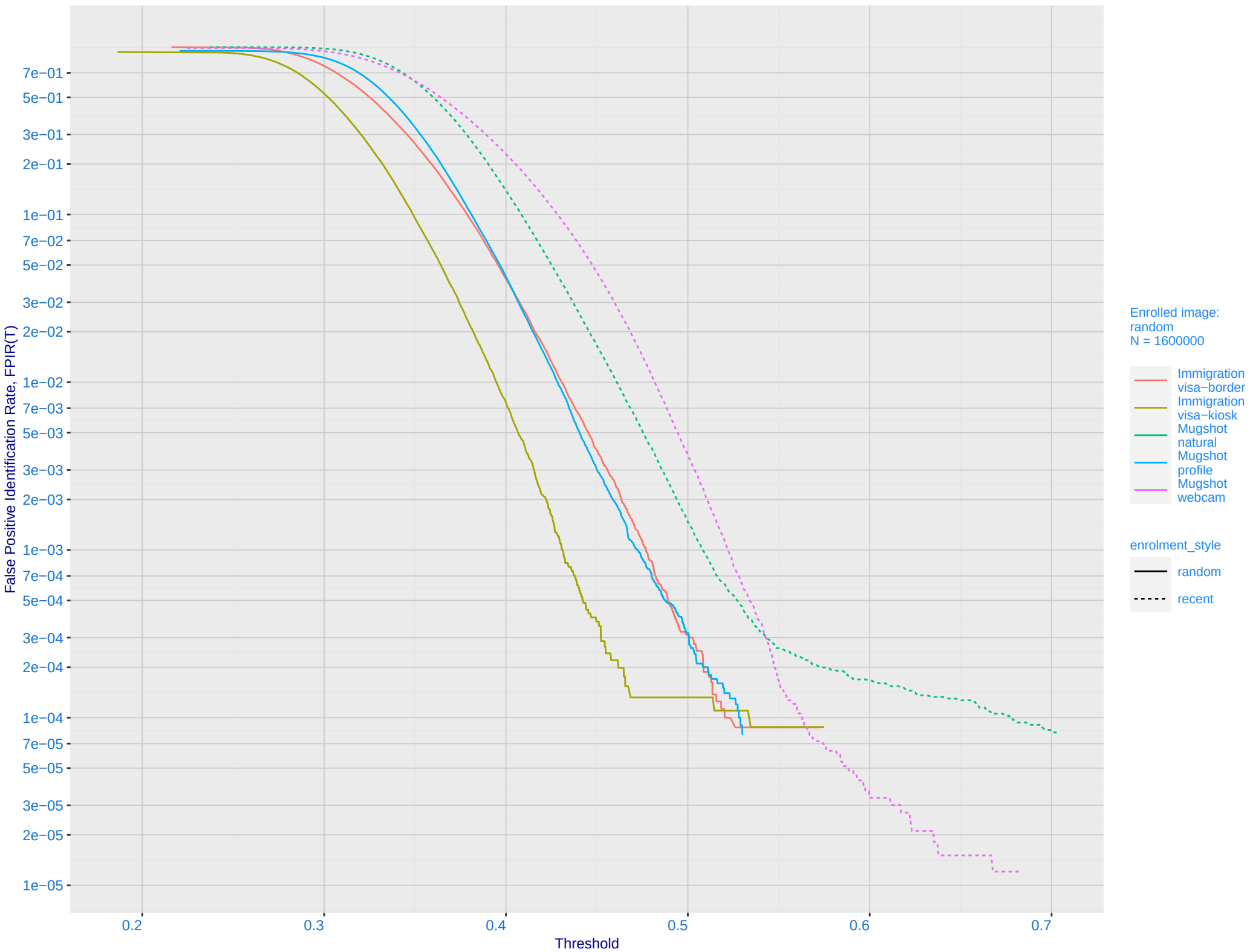
E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images



F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate

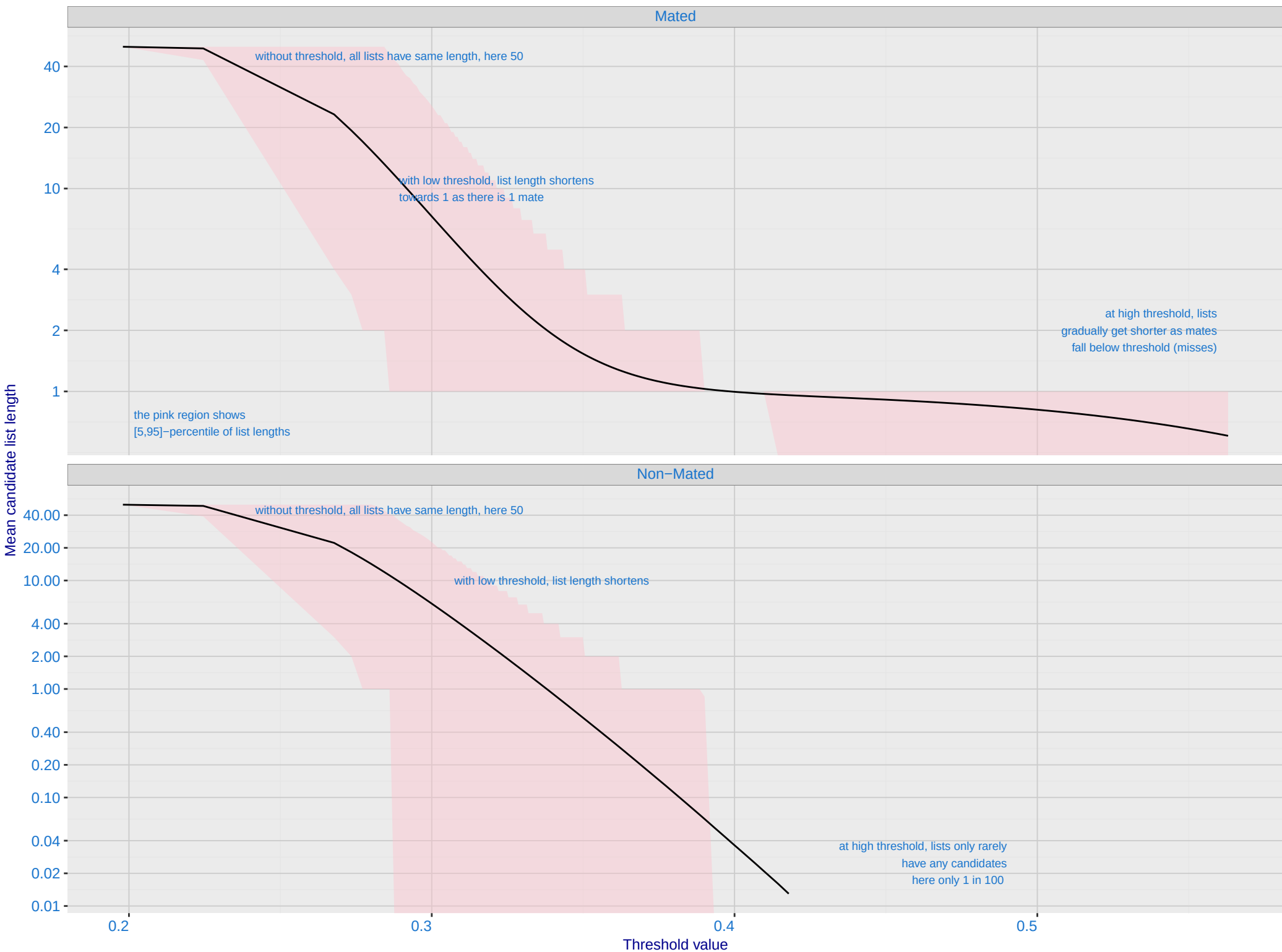


G: FPIR dependence on T by probe type for N = 1600000 subjects



H: Reduced length candidate lists for human review

Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image



I: FNIR and FPIR dependence on threshold

Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image

False {Negative | Positive} Identification Rate

FNIR(T)

0.40

0.20

0.10

0.04

0.02

0.01

FPIR(T)

1e+00

4e-01

2e-01

1e-01

4e-02

2e-02

1e-02

4e-03

2e-03

1e-03

4e-04

2e-04

1e-04

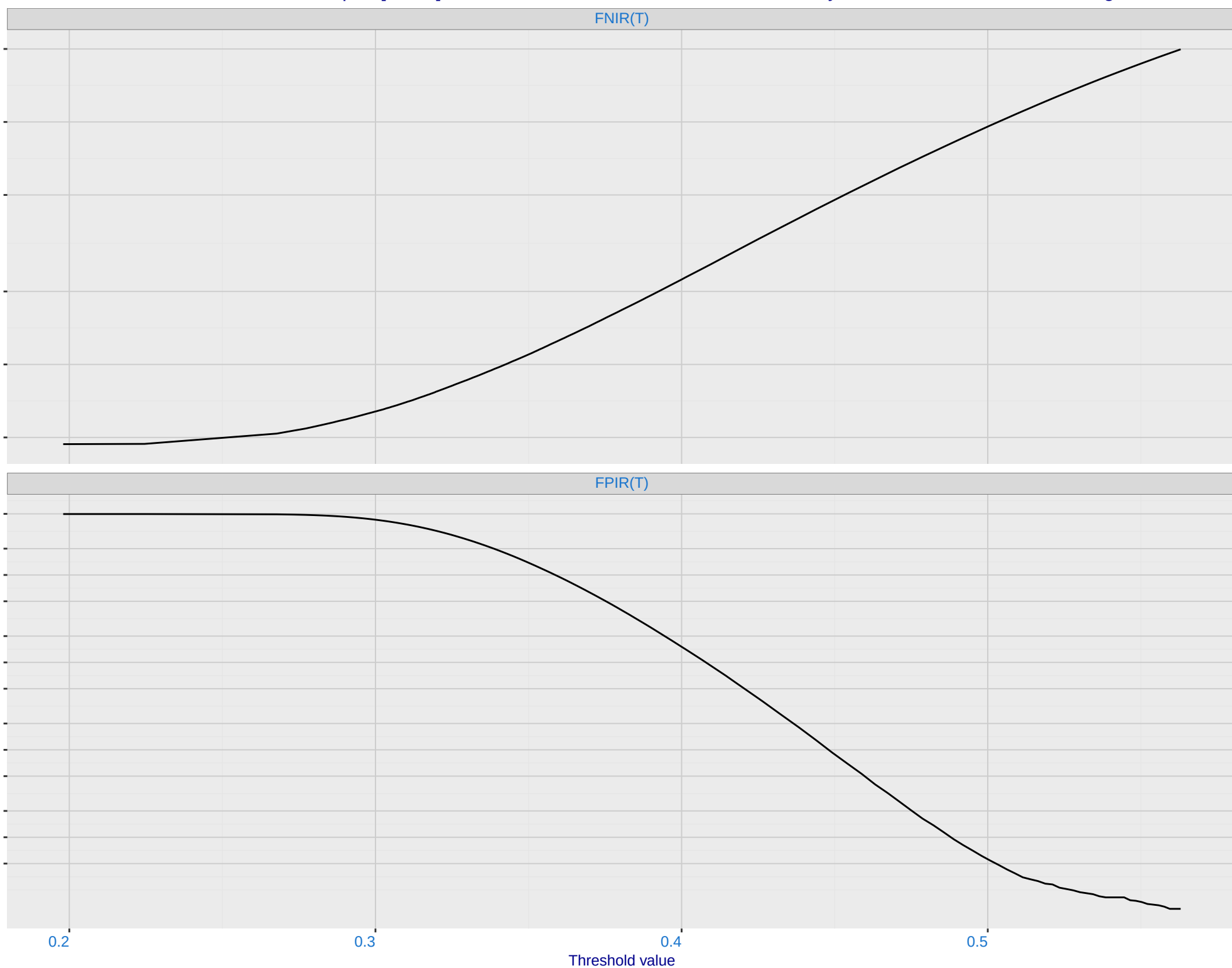
0.2

0.3

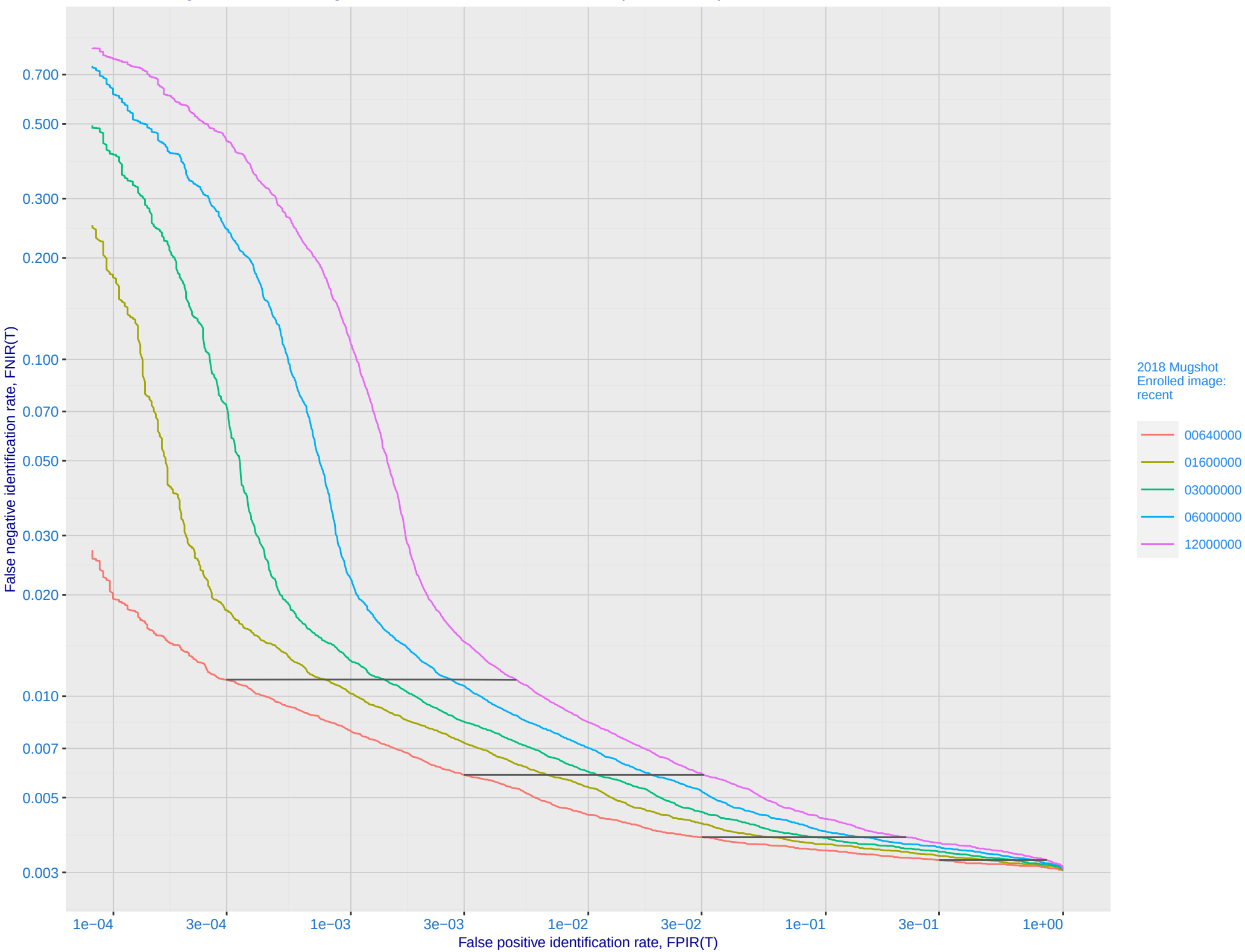
0.4

0.5

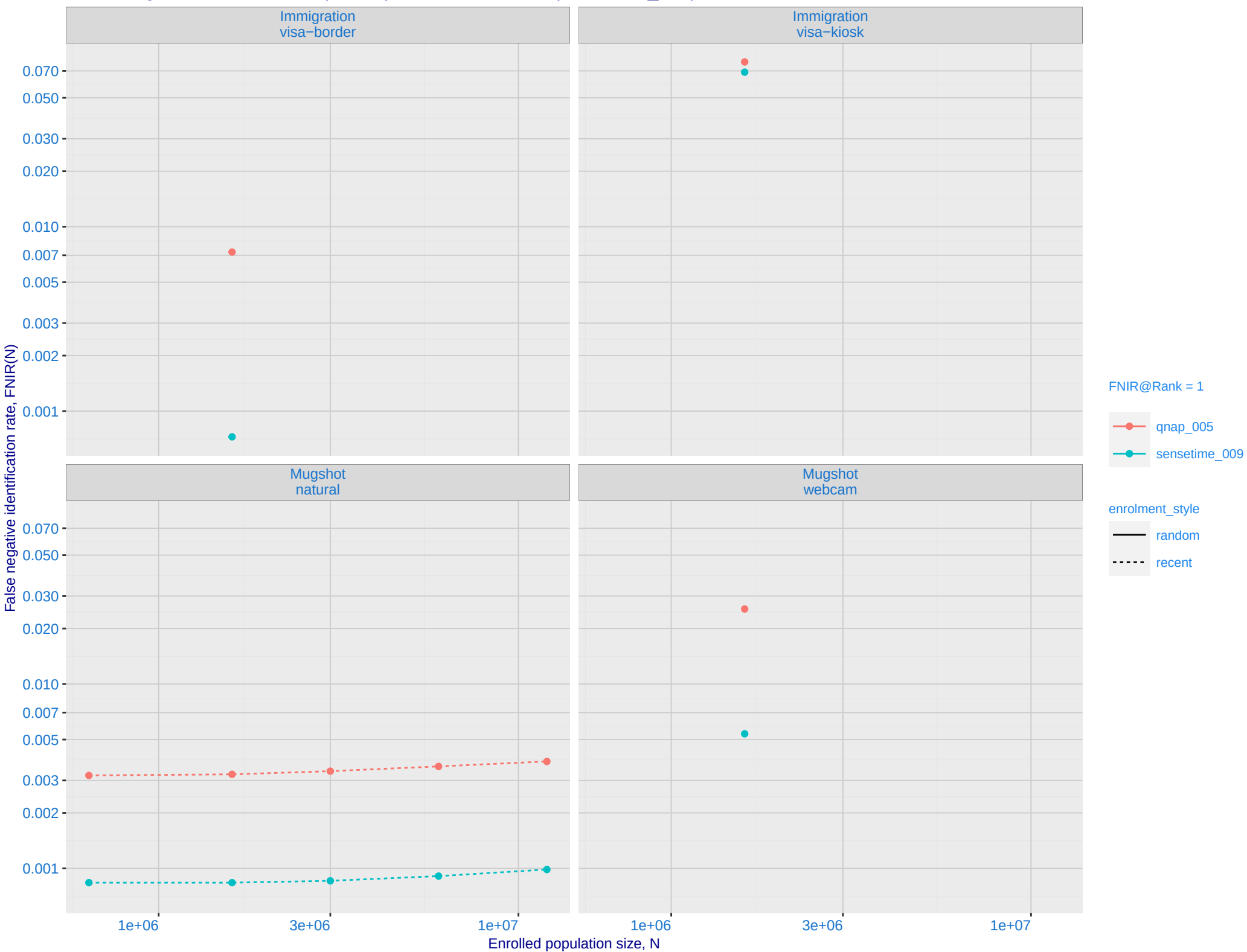
Threshold value



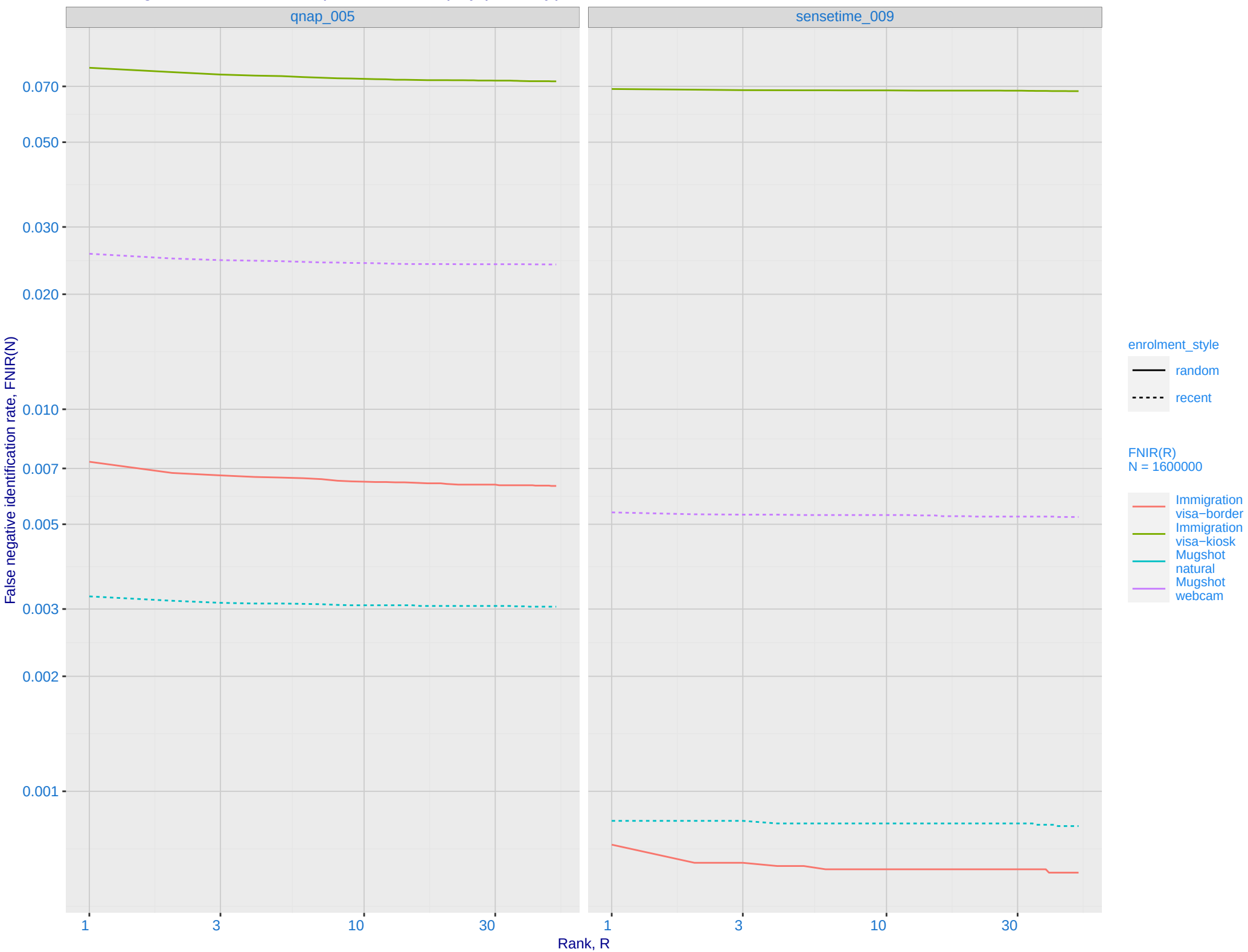
J: DET for Mugshot natural images and various N. Links connect points of equal threshold.



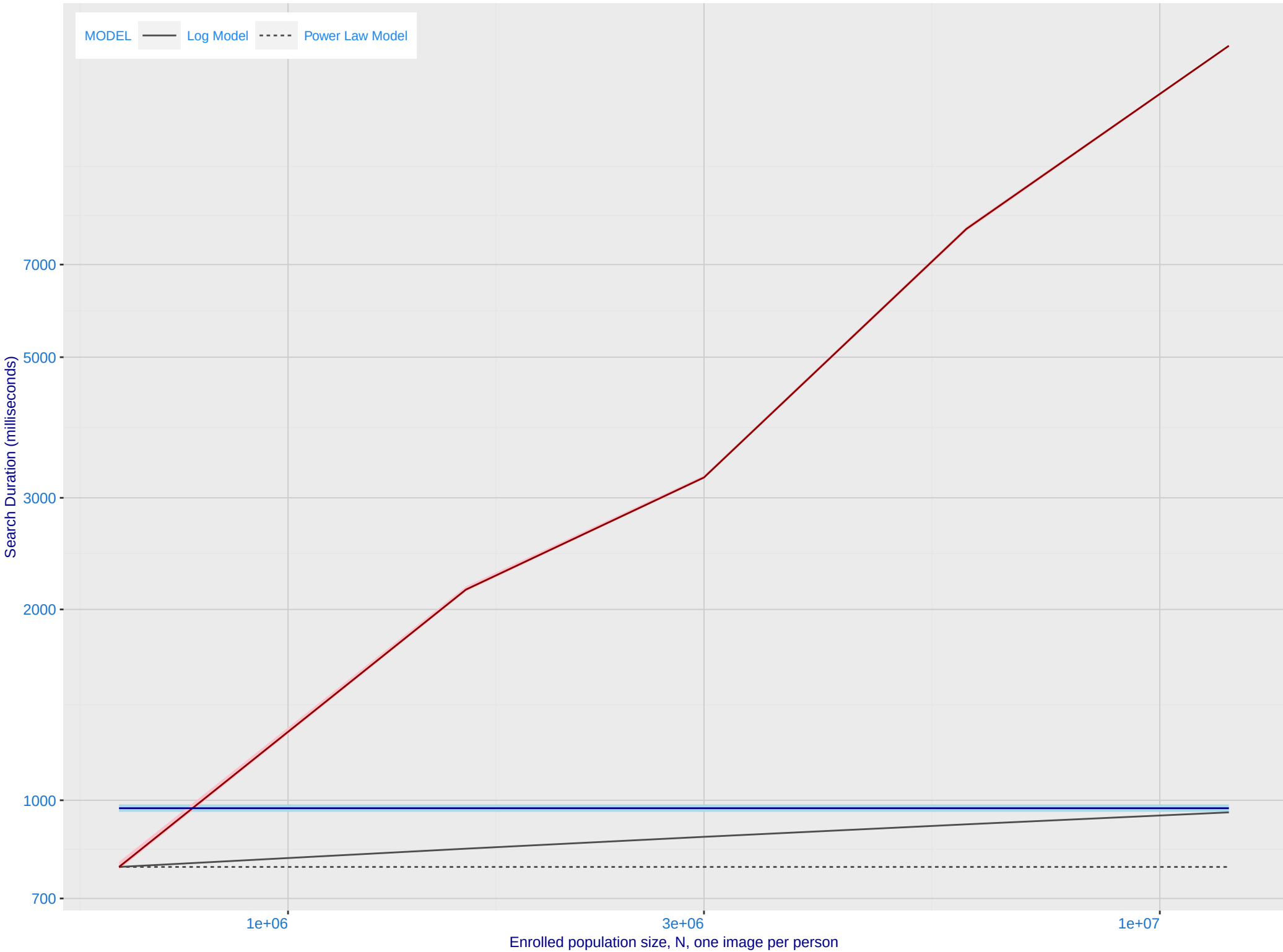
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)



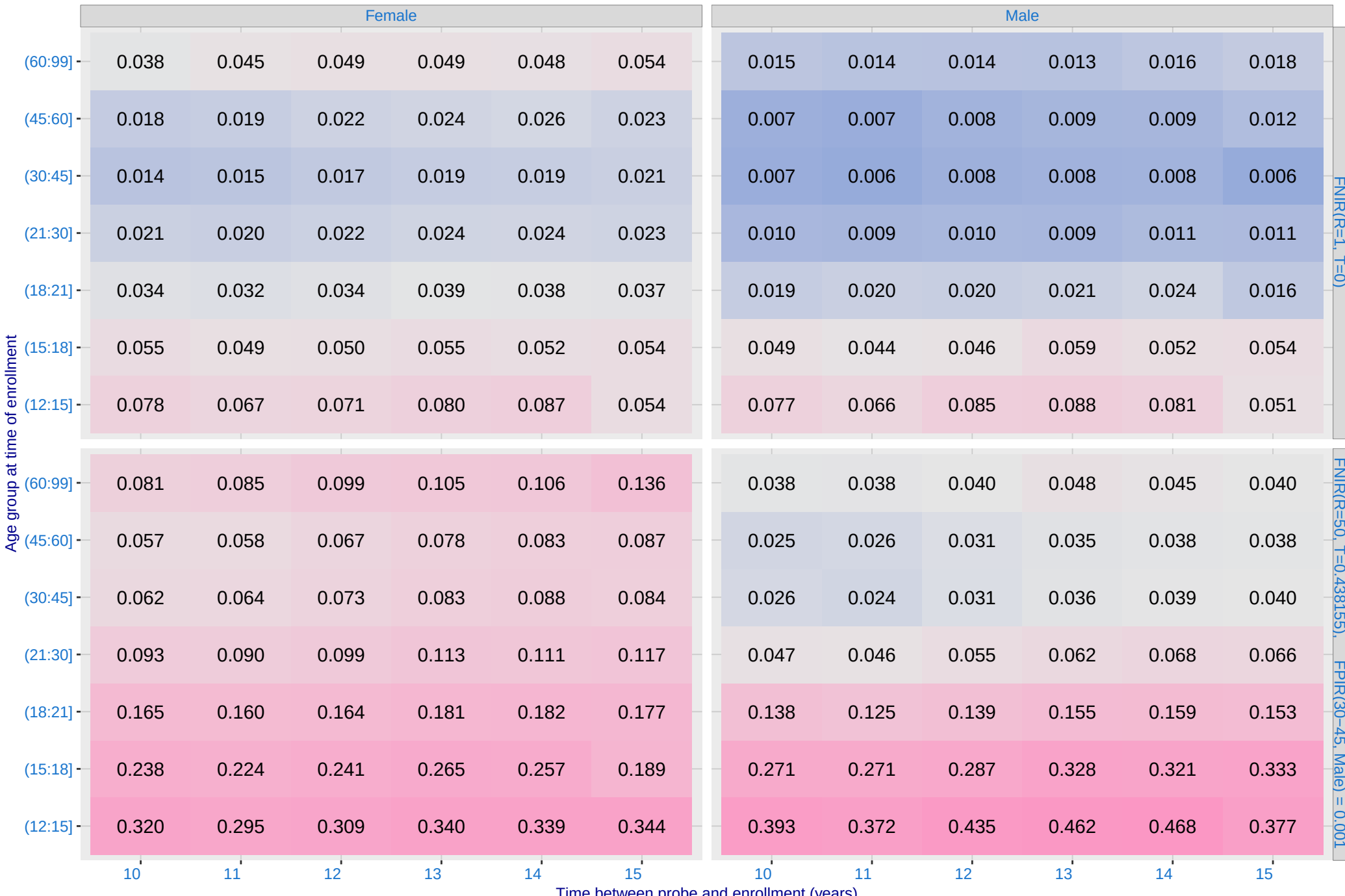
L: Investigational mode: FNIR(1600000, R, 0) by probe type



M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements. The template generation time is independent of N. The log and power-law models are fit to the first two (N,T) observations



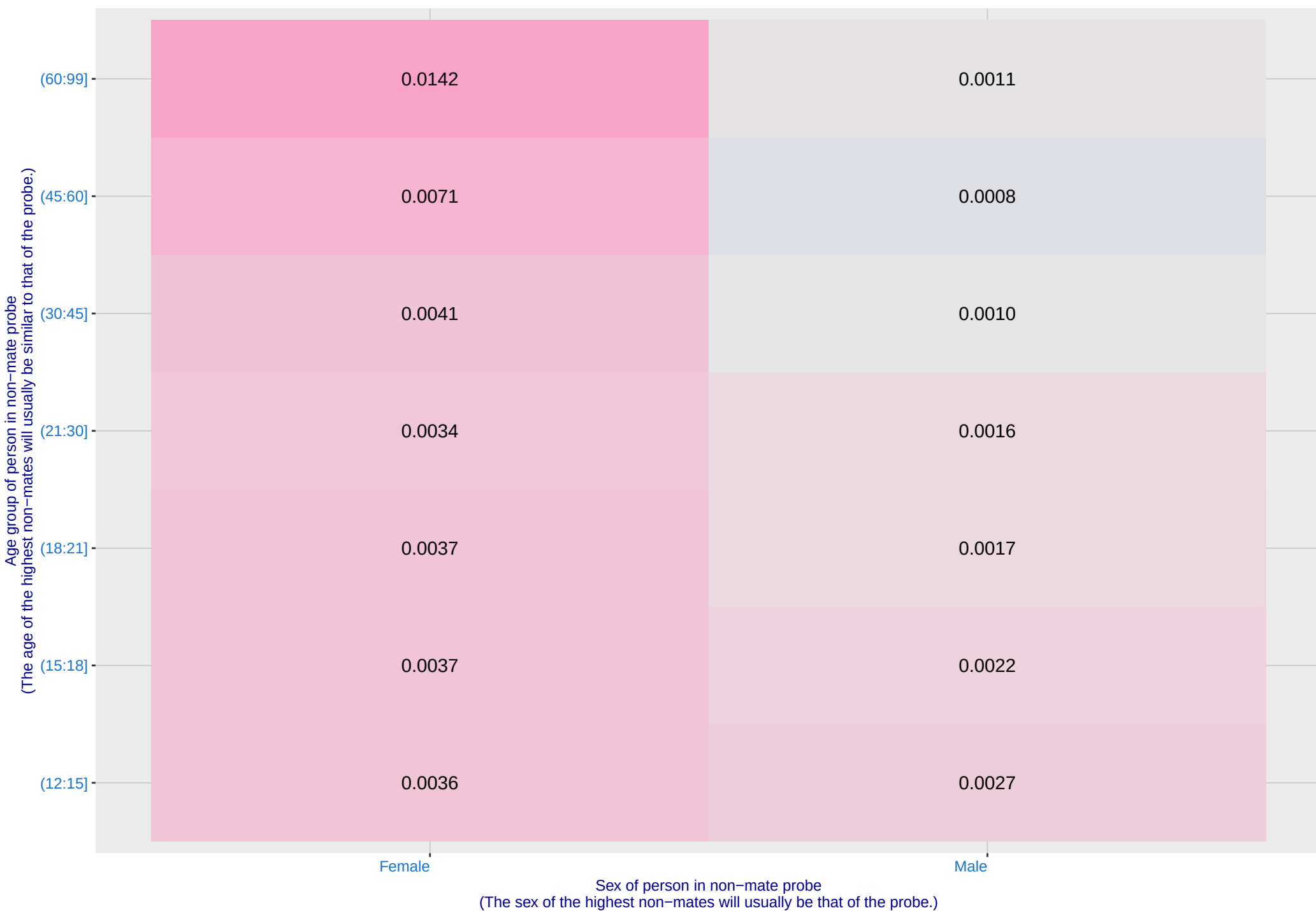
O: FNIR(T, N = 1.6 million) by sex, age and time-lapse. The top row gives investigational rank-1 miss rates. The bottom panels give high threshold for more lights-out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.

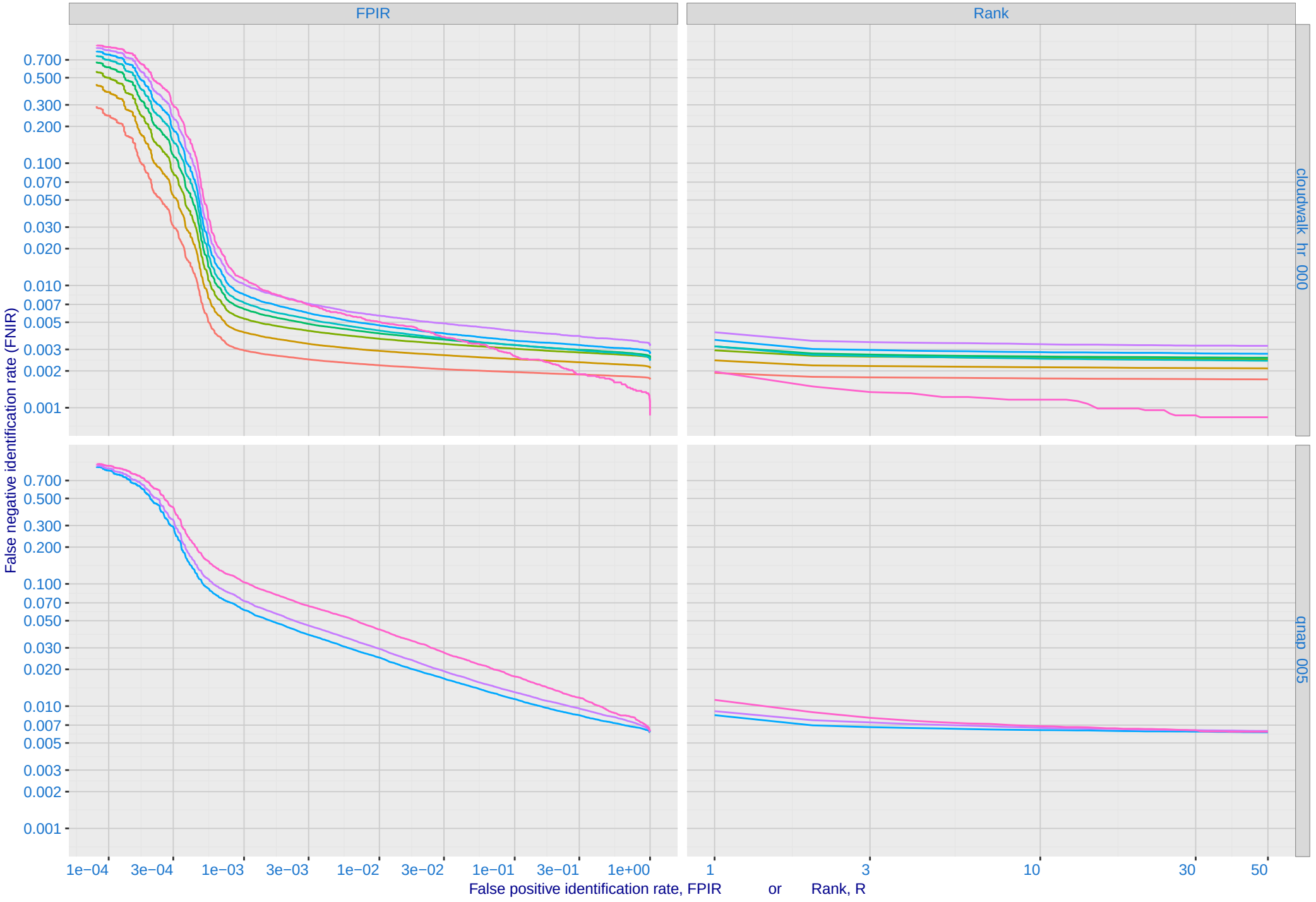
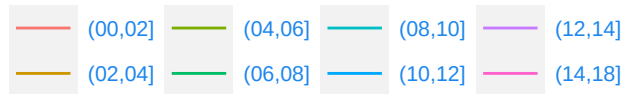
Algorithm: qnap_005, Dataset: Border-Crossing Ageing
Threshold: 0.438155 set to achieve FPIR(30-45, Male) = 0.001

Color encodes log(FPIR)



Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing

Dataset: 2018 Mugshot N = 3068801



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines

